

Wild-fire risk prediction model:

Train a model to predict the wildfire risk zones. Use the weather data including humidity, sunshine time, temperature and so on together with the fire data in different regions to predict the fire. Then use an LLM to generate English risk reports for local communities to prevent the fire.

Datasets:

Fire data:

NASA FIRMS (active fire)

firms.modaps.eosdis.nasa.gov

https://nrt3.modaps.eosdis.nasa.gov/archive/FIRMS/modis-c6.1/USA_contiguous_and_Hawaii/

Weather data:

[NOAA Public CMIP6 Dataset on Google Cloud Marketplace](#)

<https://console.cloud.google.com/marketplace/product/noaa-public/cmip6?project=fluid-script-453604-u5>

Conceptual map:

1. Key question:
 - a. What to predict: the likelihood of fire taking place in different regions in the US/ Canada
 - b. Who is the user: communities
 - c. How to measure the model: Recall, F1 score, LLM text readability
2. Dataset: listed above
3. EDA:
 - a. Analysis correlation between weather data including temperature, humidity, sunshine and wind with fire data
 - b. Temporal patterns: seasonal fire cycles
4. ML Models:
 - a. XGBoost: good for weather data
 - b. Random Forest
 - c. LSTM

5. LLM: To be decided